CSS 1

1. What is CSS?

CSS = Cascading Style Sheets

It is a language used to **style HTML pages**.

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2. CSS Rule Structure (Syntax)

Every CSS rule follows this structure:

```
selector {
    property: value;
}
```

Example:

```
p {
  color: blue;
  font-size: 30px;
}
```

Meaning:

Part	Meaning
selector	which HTML element
property	what we change
value	how we change it

---

3. Example

HTML:

```
<p>This is my paragraph</p>
```

CSS:

```
p {
  color: blue;
  font-size: 30px;
}
```

Result:

- text color → blue
- text size → 30px

---

4. CSS Selectors

Selectors are used to **choose which HTML elements will receive styles**.

Your notes mention 3 important selectors.

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1. Element Selector (Normal Selector)

Targets **all elements of a type**.

Example:

```
p {
  color: red;
}
```

All paragraphs become red.

---

2. Class Selector

Targets elements with a **class attribute**.

HTML:

```html
<p class="one">This is paragraph</p>
```

CSS:

```css
.one {
  color: red;
}
```

---

3. ID Selector

Targets **one specific element**.

HTML:

```html
<p id="title">Hello</p>
```

CSS:

```css
#title {
  color: blue;
}
```

---

5. Where Can We Write CSS?

Your notes show **three places**.

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1. Inline Style

CSS is written **inside the HTML element**.

Example:

```html
<p style="color:red; font-size:20px;">Hello</p>
```

Pros:

- quick

Cons:

- bad practice for large projects

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2. Internal Style (Internal Style Sheet)

CSS is written **inside the HTML file** inside `<style>`.

Example:

```
<head>
<style>
p {
  color: red;
}
</style>
</head>
```

Used for **single page styling**.

---

3. External Style Sheet (Best Practice)

CSS is written in a **separate file**.

Example:

```
style.css
```

Inside:

```
p {
  color: green;
}
```

Then linked in HTML:

```html
<link rel="stylesheet" href="style.css">
```

Advantages:

- clean code
- reusable
- easier maintenance

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6. Important Idea: Cascading

CSS means **Cascading Style Sheets**.

This means **styles can override each other**.

Priority (highest → lowest):

1. Inline CSS
2. Internal CSS
3. External CSS

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7. Rules to remember (CSS class identifiers)

1. Valid CSS Class Names

- Must start with **a letter, `_` (underscore), or (hyphen)**
- Can contain **letters, numbers, `_`, and**
- **Cannot start with a number**
- **Avoid special characters** like `@`, `#`, `!`, `$`

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2. Best Practice for Naming

- Use **lowercase letters**
- Prefer **kebab-case** → `user-name`

- Keep names **clear and readable**
- Avoid **ALL CAPS**
- Avoid **random long words without separation**

Good style:

`user-name`, `user-profile`, `main-container`

CSS 2

1. What is CSS Background?

The **background** in CSS controls what appears **behind an element**.

Example:

```
<div>Hello</div>
```

We can change the background of this `div`.

Example CSS:

```
div {
  background-color: red;
}
```

Result:

The `div` has a **red background**.

---

2. Ways to write colors in CSS

Your notes show **4 common formats**.

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1. Color Name

Example:

```
background-color: red;
background-color: blue;
background-color: green;
```

Simple but limited.

---

2. HEX Color

Example:

```
background-color: #FF0000;
```

Structure:

```
#RRGGBB
```

---

3. RGB

Structure:

```
rgb(red, green, blue)
```

Values go from **0** → **255**

Example:

```
background-color: rgb(147, 38, 38);
```

Example:

```
rgb(255,0,0) → red
rgb(0,255,0) → green
rgb(0,0,255) → blue
```

---

4. RGBA (with transparency)

Same as RGB but with **alpha (opacity)**.

Structure:

```
rgba(red, green, blue, alpha)
```

Alpha:

```
0 = transparent
1 = fully visible
```

Example:

```
background-color: rgba(147, 38, 38, 0.6);
```

---

3. Background Image

Property:

```
background-image
```

Example:

```
div {
  background-image: url("icon.png");
}
```

This places an **image as background**.

Example with folders

```
div {
  background-image: url("../images/icon.png");
}
```

Explanation:

```
../ = go one folder back
```

Example project structure:

```
project
 ├── css
 │    └── style.css
 └── images
      └── icon.png
```

---

4. Background Repeat

By default, background images **repeat**.

Property:

```
background-repeat
```

Options:

Value	Meaning
repeat	repeat horizontally and vertically
repeat-x	repeat horizontally
repeat-y	repeat vertically
no-repeat	show image once

Example:

```
div {
  background-repeat: no-repeat;
}
```

---

5. Background Attachment

Property:

```
background-attachment
```

Controls **how the background behaves when scrolling**.

Options:

Value	Meaning
scroll	image moves with the page
fixed	image stays fixed

Example:

```
div {
  background-attachment: fixed;
}
```

Result:

When scrolling the page, the background **does not move**.

---

6. Background Position

Property:

```
background-position
```

Controls **where the image appears**.

Example:

```
div {
  background-position: left top;
}
```

Possible positions:

Horizontal	Vertical
left	top
center	center

Horizontal	Vertical
right	bottom

You can also use percentages:

```
background-position: 40% 60%;
```

---

7. Combining Background Properties

You can combine everything:

```
div {
  background: #000 url("../images/icon.png") no-repeat scroll
center;
}
```

Meaning:

- color → black
- image → icon.png
- repeat → no-repeat
- attachment → scroll
- position → center

---

8. CSS Comments

CSS comments are written like this:

```
/* this is a comment */
```

Example:

```
/* Div style */
div {
  width: 500px;
  background: #DDD;
}
```

Comments help explain code.